

BREST BRETAGNE AP, FranceWMO: **071100**Lat: **48.4441N**Lon: **4.4120W**Elev: **99**StdP: **100.14**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.4	0.0	-4.7	2.6	0.4	-3.1	3.0	1.7	14.3	9.6	12.6	9.5	2.6	80	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.5	25.8	18.9	23.7	18.0	22.0	17.1	19.9	24.3	18.8	22.0	17.9	20.4	4.0	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.3	13.3	21.0	17.6	12.8	19.9	16.9	12.2	18.9	57.1	24.4	53.5	22.1	50.9	20.4	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	10.2	9.1	-3.9	29.6	1.6	2.4	-5.0	31.4	-6.0	32.7	-6.9	34.1	-8.1	35.8		
(5)			-4.5	21.7	1.5	1.5	-5.6	22.8	-6.5	23.7	-7.4	24.5	-8.5	25.6	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	7.0	7.1	8.4	9.8	12.5	15.1	16.9	16.8	15.2	12.8	9.7	7.6	(6)	(7)
	DBStd	4.44	3.01	2.73	2.62	2.56	2.57	2.43	2.10	2.05	2.32	2.63	2.57	3.15	(7)	(8)
	HDD10.0	412	97	86	62	33	6	0	0	0	0	6	36	86	(8)	(9)
	HDD18.3	2488	350	316	308	256	181	101	56	57	99	171	261	333	(9)	(10)
	CDD10.0	994	4	3	13	27	84	154	212	211	155	94	26	11	(10)	(11)
	CDD18.3	30	0	0	0	0	1	5	11	9	4	0	0	0	(11)	(12)
	CDH23.3	216	0	0	0	1	11	51	78	52	22	2	0	0	(12)	(13)
	CDH26.7	37	0	0	0	0	1	11	12	11	2	0	0	0	(13)	(14)
(14) Wind	WSAvg	4.3	4.9	4.8	4.7	4.4	4.2	4.0	3.9	3.7	3.8	4.2	4.5	4.8	(14)	(15)
(15) Precipitation	PrecAvg	1161	140	112	95	82	76	58	59	68	80	118	135	142	(15)	(16)
	PrecMax	1606	287	266	221	221	161	159	153	163	169	255	279	331	(16)	(17)
	PrecMin	784	19	10	14	5	11	2	10	10	11	10	33	22	(17)	(18)
	PrecStd	181	67	57	50	50	39	37	35	41	43	57	62	68	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	14.9	19.2	22.3	25.0	28.3	28.1	27.9	25.9	21.3	16.5	14.2	(19)	(20)
		MCWB	11.8	10.8	12.6	14.4	17.2	20.0	20.3	20.2	19.0	17.1	14.4	12.7	(20)	(21)
	2%	DB	12.4	12.4	15.4	18.4	21.4	24.4	25.5	24.2	23.1	19.0	15.1	13.5	(21)	(22)
		MCWB	11.4	10.4	11.1	13.0	15.6	17.9	18.9	19.1	17.8	15.9	13.7	12.4	(22)	(23)
	5%	DB	11.8	11.6	13.2	15.6	19.1	21.8	23.3	22.1	20.9	17.6	14.2	12.8	(23)	(24)
		MCWB	11.0	10.3	10.5	11.6	14.7	16.7	17.8	18.0	16.7	15.5	13.0	11.9	(24)	(25)
	10%	DB	11.2	11.0	12.0	13.8	17.1	19.7	21.2	20.5	19.2	16.6	13.6	12.2	(25)	(26)
		MCWB	10.2	9.9	10.3	10.8	13.6	15.6	16.9	17.0	16.0	15.0	12.4	11.3	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.5	12.1	13.2	15.4	18.1	20.8	21.1	21.5	20.1	17.7	15.3	13.4	(27)	(28)
		MCDB	12.8	13.1	17.4	20.0	22.6	26.8	26.5	25.7	24.5	19.9	15.7	13.7	(28)	(29)
	2%	WB	11.8	11.3	12.1	13.4	16.3	18.7	19.6	19.7	18.5	16.8	14.2	12.7	(29)	(30)
		MCDB	12.3	11.9	13.9	17.2	20.3	22.9	24.3	23.0	21.3	18.1	14.8	13.2	(30)	(31)
	5%	WB	11.2	10.7	11.4	12.2	15.1	17.2	18.4	18.6	17.6	16.0	13.4	12.1	(31)	(32)
		MCDB	11.7	11.4	12.7	14.6	18.5	20.8	21.8	21.1	19.6	17.2	14.1	12.7	(32)	(33)
	10%	WB	10.5	10.1	10.7	11.5	14.0	16.1	17.4	17.8	16.8	15.3	12.6	11.5	(33)	(34)
		MCDB	11.1	10.9	11.8	13.3	16.4	18.8	20.1	19.6	18.5	16.4	13.4	12.1	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.6	6.4	7.3	7.5	7.6	7.5	7.5	7.8	6.5	5.4	5.1	(35)	(36)
		MCDBR	4.4	6.0	8.5	10.8	11.1	11.9	11.8	10.2	10.5	7.4	5.5	4.2	(36)	(37)
	5% WB	MCWBR	4.3	4.8	5.5	6.3	6.5	6.4	6.1	5.6	5.8	4.9	4.5	4.1	(37)	(38)
		MCWBR	4.1	4.9	6.8	8.7	9.9	10.3	9.9	8.5	8.6	6.1	4.9	4.2	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.334	0.348	0.373	0.381	0.385	0.378	0.379	0.368	0.358	0.355	0.335	0.327	(39)	(40)
		taud	2.396	2.379	2.324	2.321	2.355	2.399	2.400	2.451	2.462	2.456	2.425	2.364	(40)	(41)
	Ebn at Noon		723	800	836	868	877	884	877	873	847	784	715	673	(41)	(42)
		Edh at Noon	69	86	106	118	118	114	112	102	92	79	66	63	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		0.96	1.77	2.95	4.62	5.48	5.93	5.63	4.83	3.85	2.26	1.21	0.80	(43)	(44)
	RadStd		0.07	0.18	0.30	0.39	0.48	0.62	0.31	0.39	0.28	0.17	0.10	0.10	(44)	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page